INCH-POUND

MIL-DTL-7830E
20 March 2007
SUPERSEDING
MIL-L-7830D
15 September 1983

DETAIL SPECIFICATION

LIGHT ASSEMBLY, MARKER, AIRCRAFT OBSTRUCTION

Inactive for new design after 19 March 2007

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an aircraft obstruction marker light assembly.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center Philadelphia, ATTN: DSCP-NASA, 700 Robbins Ave, Philadelphia, PA 19111-5096 or emailed to dscpg&ispeccomments@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/

AMSC N/A FSC 6210

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specifications For
MIL-DTL-7989	Cover, Light – Transmitting, For Aeronautical Lights, General Specifications For
AN2541	Lens, Aircraft Obstruction Marker Light
AN2547	Light, Marker, Aircraft Obstruction

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements For

(Copies of this document are available from http://www.sae.org or the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Components. The light assembly shall consist of a lens and a light , as specified by the procuring activity. Each item supplied will be specified by its AN part number in the invitation for bids.

3.3 Materials. Materials shall conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications shall be of the best commercial quality and suitable for the purpose intended.

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3.3.1 Metals. Metals shall be of the corrosion-resistant type, unless protected to resist corrosion during normal service life.

3.4 Design and construction.

3.4.1 Lens. The dimensions of the lens shall be shown on AN2541. The type and color shall be in accordance with the AN part number specified (see 6.2).

3.4.1.1 Glass. The glass shall be class B conforming to MIL-DTL-7989.

3.4.1.2 Colors. The colors shall be type I, grade D, conforming to SAE-AS25050.

3.4.1.3 Optical design. The optical design of the lens and necessary fittings shall be such that, when the lens is in place on a light (AN2547-1) equipped with a 325 lumen series lamp operated at rated lumens, the equivalent clear lens vertical candlepower at any angle will equal or exceed the values shown on figures 1 and 2 for the corresponding angle. The equivalent clear lens candlepower shall be determined by dividing the measured candlepower by the transmission ratio for the color being measured as follows:

Transmission ratio

Color	Ratio
Clear	1.00
Yellow	0.40
Green	0.15
Red	0.13
Blue	0.008

3.4.2 Light. The light shall conform to AN2547 and shall be designed to accommodate a lens conforming to AN2541. The light shall be made of hydroformed sandcast, or die cast nonferrous corrosion-resistant metal or a ferrous metal protected by an approved commercial corrosion-resistant finish such as galvanizing, sheradizing, or baked enamel. The light shall house the socket and hold it rigidly in such a manner as to place the lamp filament at the proper position. The light shall be of separate construction to permit relamping without unscrewing the lens and shall be weatherproof. The bottom section of the light shall be appropriately grooved to provide adequate purchase for the setscrew in the top section. Drain holes shall be drilled in the light as shown on AN2547.

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3.4.2.1 Socket assembly. The socket assembly shall be supplied as a part of the light. The type and number required will be determined by the AN part number of the light. Sockets and receptacles shall be manufactured of glazed porcelain, or other material approved by the procuring activity, and nonferrous corrosion-resistant metal. The socket assembly receptacle, mounting, and light shall be such as to accurately place and hold the light center of the lamp within 7/32 inch of the vertical axis of the light center, and within 5/64 inch above or below the correct light center. This tolerance shall include all allowable tolerances in the light, socket, receptacle, and mountings. The socket, socket receptacle, socket mounting and mounting screws shall be such that, when these parts are correctly assembled, irrespective of their location to each other due to manufacturing tolerances, the light center tolerances specified shall not be exceeded. Both the series and multiple socket assemblies shall be so designed that when a lamp is installed, the plane of the filament will be located with respect to the mounting holes as shown on AN2547.

3.4.2.1.1 Multiple sockets. Sockets for multiple service shall be of the medium prefocus type. The sockets shall be in accordance with AN2547. The medium prefocus socket shall be positioned for use with a medium prefocus lamp having a 2-3/4 inch light center length. When the multiple socket designated by AN2547-10 is specified, it shall be furnished complete with two 4-36 mounting screws of the proper length and washers for assembling the socket to the mounting.

3.4.2.1.2 Series socket assembly. The socket assembly for use on series circuits shall consist of a medium prefocus-type socket and receptacle in accordance with AN2547. The socket assembly shall be furnished complete with two 4-36 mounting screws of the proper length and washers for assembling the socket to the mounting. An easily replaceable disk film cutout not greater than 25/32 inch in diameter and not more than 0.07 inch in thickness shall short circuit the socket in case of lamp failure. The disk cutout shall have breakdown voltage of not less than 20V nor more than 100V. The receptacle shall become automatically short-circuited during removal of the socket and before the socket loses contact with the receptacle. The design shall be such that line binding posts are not exposed during the relamping process.

3.4.2.2 Setscrews. The setscrews shall be wing or knurled thumbscrews.

3.4.2.3 Chain. Unless otherwise specified, a chain, linking the top and bottom sections, shall be supplied. The chain shall be of such length that the top section be removed with a lamp in place.

3.5 Operating conditions. The light assembly shall be designed and constructed for continuous outdoor service under all weather conditions.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

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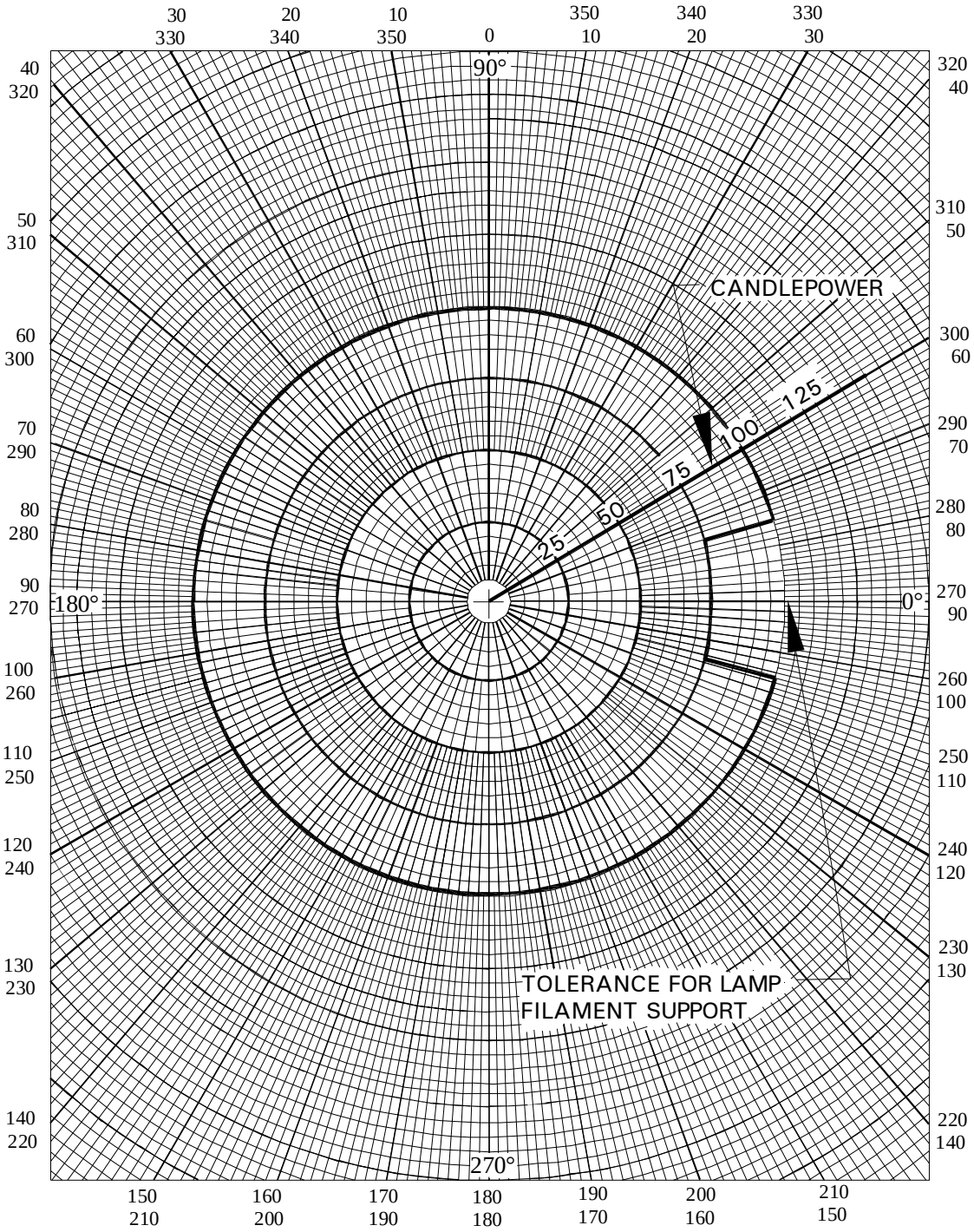


FIGURE 1. Horizontal distribution.

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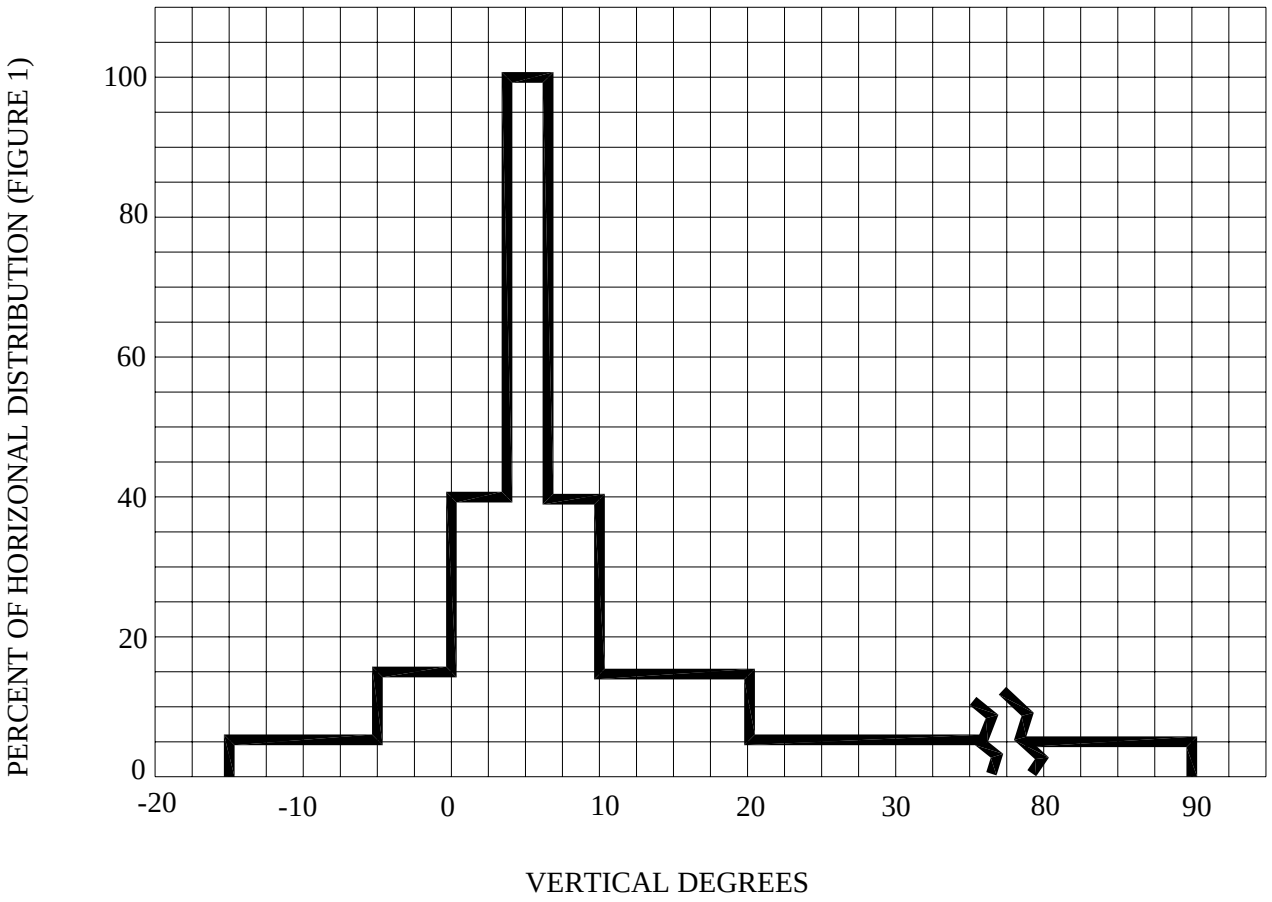


FIGURE 2. Vertical distribution through any vertical plane

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3.7 Threads.

3.7.1 Screw. Screw threads shall conform to MIL-S-7742, class 2.

3.7.2 Pipe. Pipe threads shall conform to FED-STD-H28.

3.8 Finish. Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.9.1 Castings. Each casting shall have the following information cast with raised letters in a position where it can be easily read:

AN Part No.
Manufacturer's name or trademark

3.10 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering this class of work.

4. VERIFICATION

4.1 Classification of inspections: The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).

4.2 First article inspection.

4.2.1 Test samples. The test samples shall consist of one complete sample of each part number representative of the production equipment.

4.2.2 Test report. Upon completion of the first article tests, a test report shall be prepared (see 6.4) and three complete copies of the report furnished to the procuring activity.

4.2.3 First article tests. First article tests shall consist of all tests described under 4.4.

4.3 Conformance inspections. The conformance inspections shall consist of:

a. Individual tests.

b. Sampling plan tests

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4.3.1 Individual tests. Each light assembly shall be subjected to the following tests as described under 4.4:

a. Examination of product.

b. Insulation test.

4.3.2 Sampling plan and tests.

4.3.2.1 Lot. A lot shall consist of light assemblies manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.3.2.2 Sampling plan. One lens, one light, and one socket assembly shall be selected at random from each 200 or fraction thereof produced and subjected to the following tests:

a. Photometric and chromaticity.

b. Dimensional checks.

c. Series socket assembly.

4.3.2.2.1 Rejection and retest. When one or more light assemblies from a Lot fail to meet the requirements of the specification, acceptance of all light assemblies in the lot will withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.2.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.3.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test methods.

4.4.1 Examination of product. The light assembly shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.4.2 Insulation. Each light assembly shall be tested for insulation integrity by applying 5,000 volts, 50 to 60 cps ac power for the series light assembly and

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1,000 volts, 50 to 60 cps ac power for the multiple service light assembly, between the lamp circuit in the socket and the fixture housing for a period of 1 minute. Breakdown of the insulation during these tests shall be cause for rejection.

4.4.3 Photometric and chromaticity. One lens and filter, if used, shall be subjected to photometric tests to determine compliance with all the photometric requirements of this specification, and to chromaticity tests to determine compliance with SAE-AS25050.

4.4.4 Dimensional checks. One light assembly shall be subjected to dimensional checks to determine compliance with the applicable drawings.

4.4.5 Series socket assemblies. One series socket assembly shall be assembled with proper light to determine that operation is satisfactory and conforms to 3.4.2.1.2.

4.4.6 Multiple socket. One multiple socket shall be assembled with proper mounting to determine that the socket is satisfactory and conforms 3.4.2.1.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The light assembly covered by this specification is intended for mounting on either poles or obstacles.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. The AN part number of the specific items desires (see 3.4.1, 3.4.2, 3.4.2.1.1, and 3.4.2.1.2).

b. Deletion or substitution for the chain specified in 3.4.2.3, if not required.

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c. Location and conditions for first article testing (see 4.2).

d. Packaging requirements (see 5.1).

6.3 Lamps. The light assembly is designed to utilize one of the following lamps (not furnished with the light assembly).

6.3.1 Multiple. A lamp of 40 or 100w, cc-2 filament (plane of filament through center of fins on base), A-21 clear bulb, 2 3/4 inch light center length, 5 5/16 inch maximum overall length, medium prefocus base.

6.3.2 Series. A lamp of 325 or 1,020 lumens, 6.6 amp, c-8 filament, A-21 clear bulb, 2 3/4 inch light center length, 5 1/16 inch maximum overall length, medium prefocus base.

6.4 Preparation of test report. The test report should be prepared in accordance with the guidance of MIL-HDBK-831.

6.5 Subject term (key word) listing.

Globe lamp
Marker lamp
Obstacle light

6.6 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army – CR4
Navy – AS
Air Force – 99

Preparing activity:
DLA – IS

(Project 6210-2006-024)

Review activity:
Navy – YD
Air Force – 11

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